IAU Offices of Astronomy for Development: Progress and plans

Jordan, 2nd December 2015

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science & technology
Department:
Science and Technology
REPUBLIC OF SOUTH AFRICA

IAU

NRF National Research Foundation | SAAO South African Astronomical Observatory

The IAU Strategic Plan 2010-2020:



CULTURE AND SOCIETY

TECHNOLOGY AND SKILLS

SCIENCE AND RESEARCH

ASTRONOMY

- **PERSPECTIVE IMMENSITY OF UNIVERSE** — Tolerance and global citizenship
- **OPTICS** — High-precision adaptive optics
- **COMPUTERS** — Fastest hardware; Complex software
- **ELECTRONICS** — Most sensitive detectors; Fastest clocks
- **HISTORY** — Evolution of Universe; Our roots
- **ANTHRO-POLOGY** — Ancient civilizations; Our roots
- **SPACE** — Satellite Miniaturization; Precision
- **INSPIRATION** — Career in science and technology
- **PHYSICS** — Laboratory of extremes; Making heavy elements
- **MATHEM-ATICS** — Abstract thought
- **BIOLOGY** — Building blocks of life
- **CHEMISTRY** — Producing organic molecules

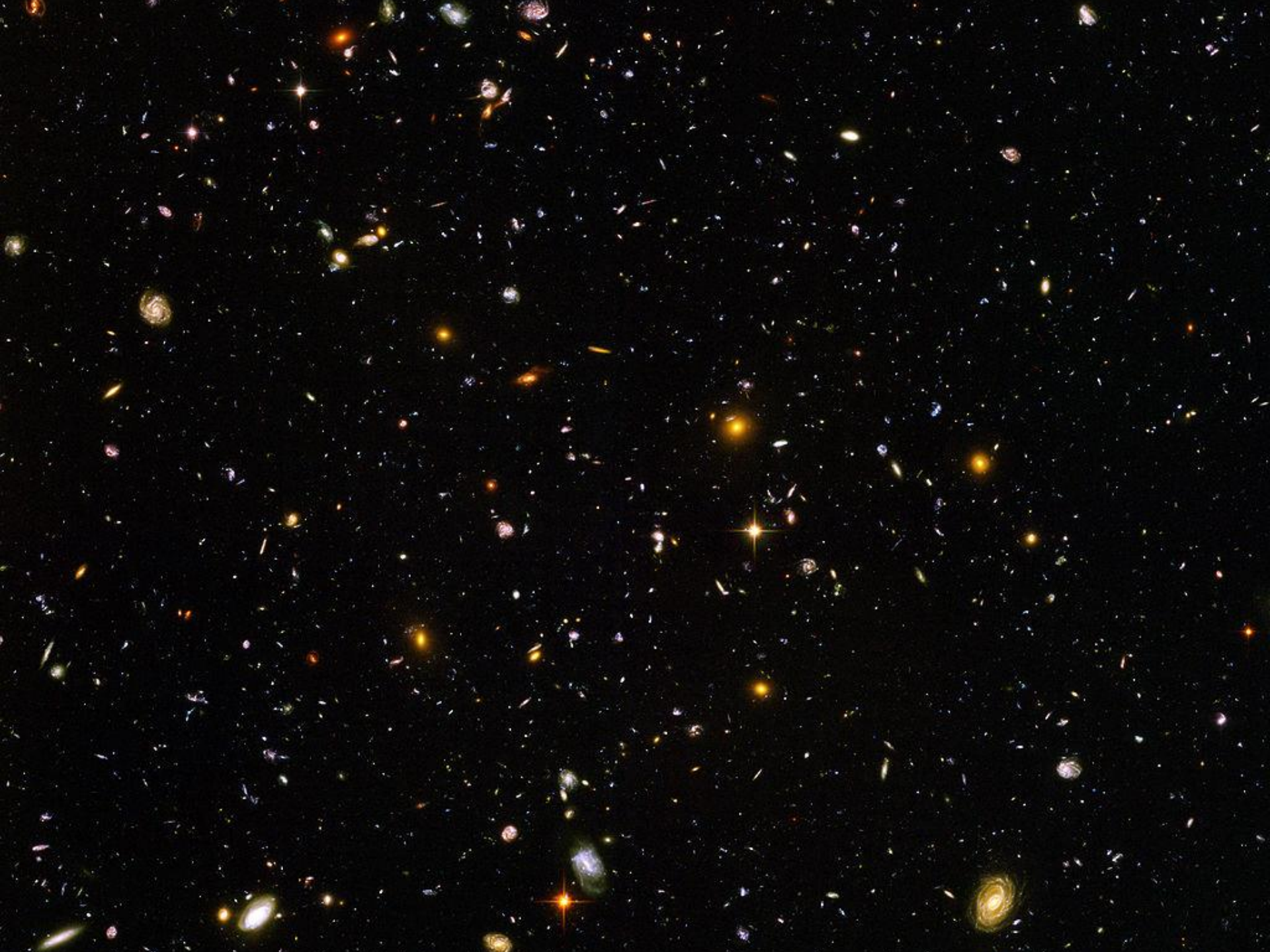
OAO - OAD - IAU (Divisions)

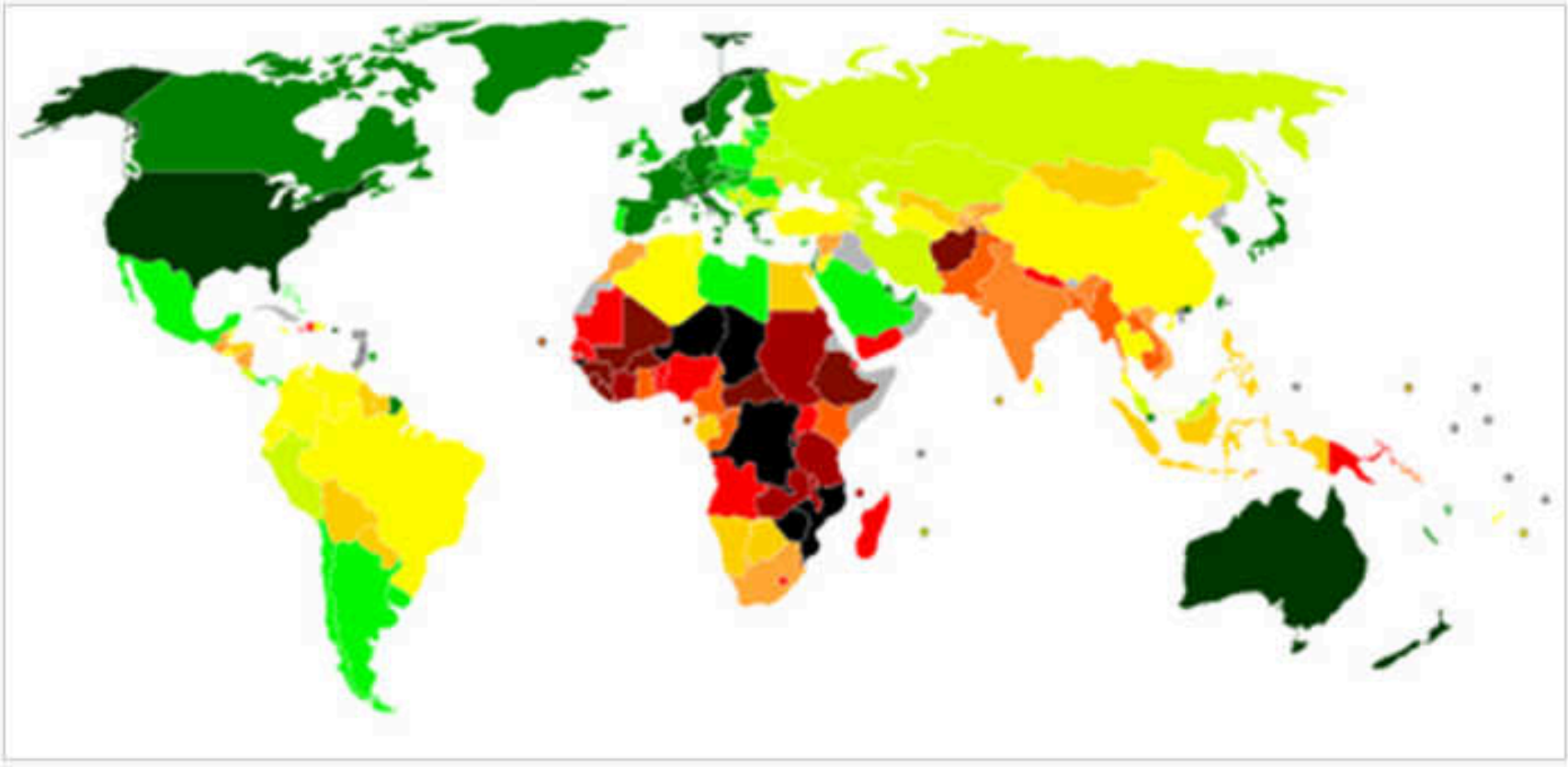
Knowledge

Access

Development

International Astronomical Union | Office of Astronomy for Development





World map indicating the Human Development Index (based on 2010 data, published on November 4, 2010)[1]

0.900 and over	0.650–0.699	0.400–0.449
0.850–0.899	0.600–0.649	0.350–0.399
0.800–0.849	0.550–0.599	0.300–0.349
0.750–0.799	0.500–0.549	under 0.300
0.700–0.749	0.450–0.499	Data unavailable

CNN

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World



Obama warns Putin

Russia shouldn't intervene in Syrian civil war, Obama says

Moscow: Turkey protecting ISIS ties

War on ISIS: Who's doing what?

Turkey: Don't 'play with fire'

BBC



US tells Turkey and Russia to end row

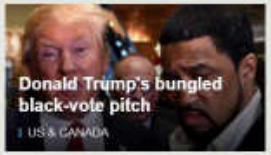
US President Barack Obama urges Turkey and Russia to calm tensions over the downing of a Russian warplane and to focus on fighting IS militants.

EUROPE



Does Turkey really get its oil from Islamic State?

EUROPE



Donald Trump's bungled black-vote pitch

US & CANADA



Faulty part 'factor' in AirAsia crash



US to deploy special force to Iraq



Blast at Istanbul metro station

ALJAZEERA

NEWS PROGRAMMES OPINION INVESTIGATIONS ABOUT video

Topics: Lebanon Nusra Front ISIL Turkey Kenya

Drowning Megacities
On the front lines of the African climate change battle

WATCH NOW
Fault Lines

TV SCHEDULE
Newshour [IN 18 MIN.]

Who's ki... clerics?

NEWS AS WE GET IT 01 DEC 2015

BLOG 14 minutes ago



US 'terror' arrests at highest level since 9/11

@AJENEWS 25 minutes ago

Images from Skala, Greece, earlier today by Al Jazeera's @P_Strickland_ who's reporting on the plight of refugees

@AJENEWS 31 minutes ago

12 rights groups call on Egypt to free investigative reporter Ismail Alexandrani by @anealla

@AJENEWS 33 minutes ago

Egypt extends detention of journalist by 15 days, charges include "spreading false news"

Updated

Iraq: We don't need foreign troops to fight ISIL



Iraqi PM's comment come after U... defence secretary said extra troop... will assist forces in fight against IS...

01 Dec 2015 22:28 GMT | War & Conflict, Mi... East, ISIL, Iraq, Syria

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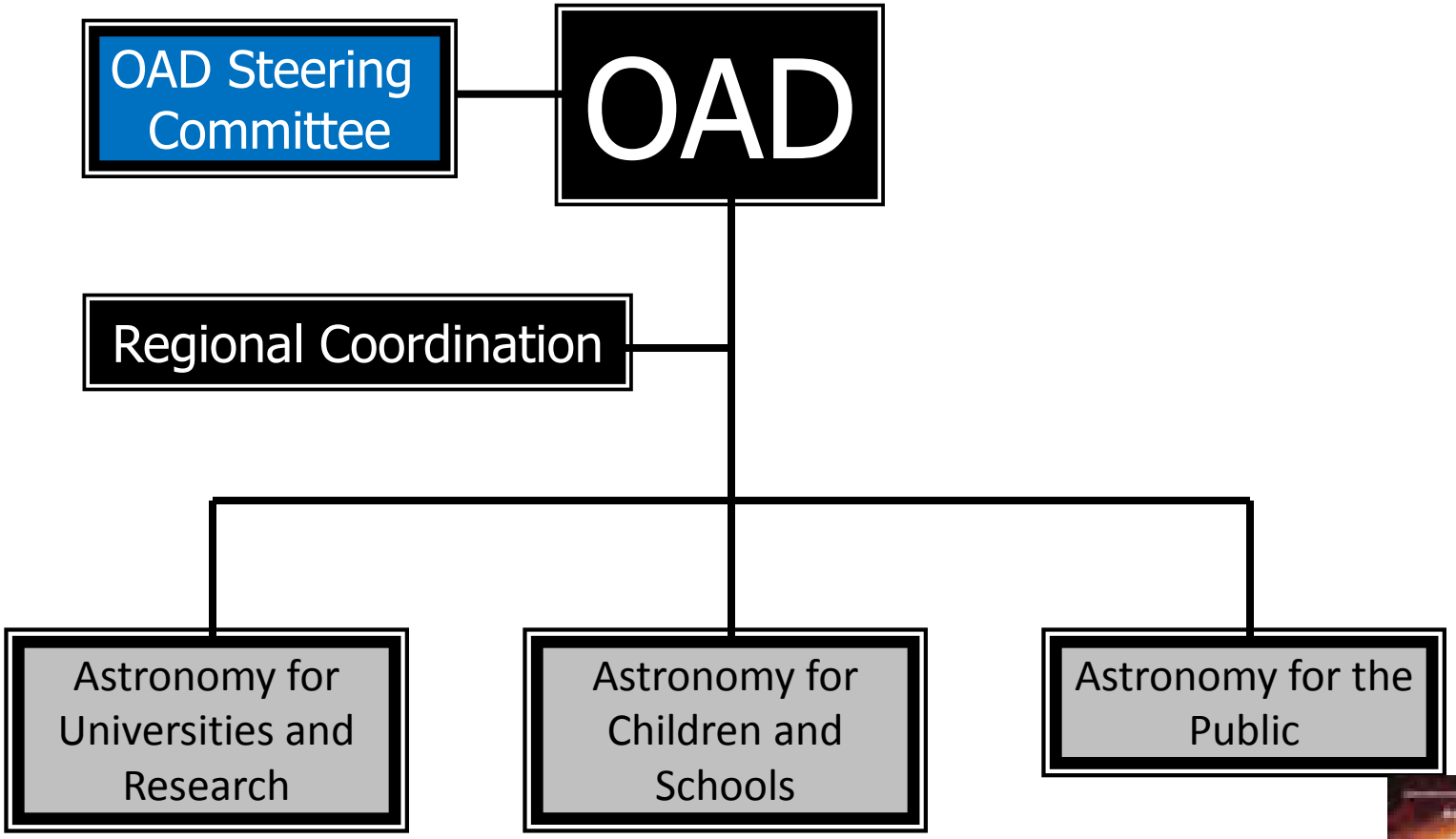
Cameron Attacks 'Terrorist Sympathiser' MPs

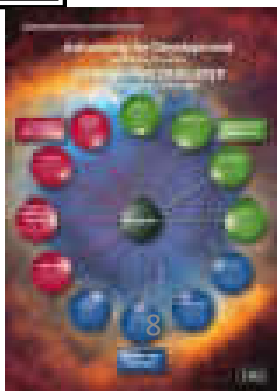


Astronomy for Sustainable Development!

- **Social benefits** (common humanity, scientific engagement & discourse)
- **Human capital development** (education, skills, career choices)
- **Economic growth** (knowledge economy, innovation)
- **Human welfare** (all of the above, technology transfer)

OAD Structure



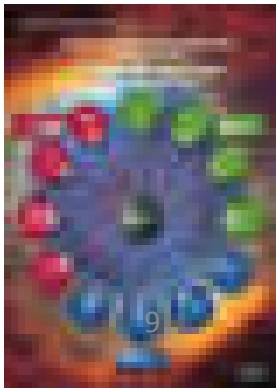


"Astronomy for a better world!"

OAD Structure

OAD Steering Committee

- Ajit Kembhavi (Chair - nominated by IAU)
- Teresa Lago (nominated by IAU)
- Katrien Kolenberg (nominated by IAU)
- Khotso Mokhele (nominated by NRF)
- Renee Kraan-Korteweg (nominated by NRF)
- Saalih Allie (nominated by NRF)



OAD Structure

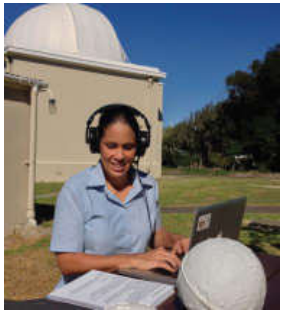


OAD

Staff:
Kevin
Eli Grant
Nuhaah








Fellows:
Silvia
Wanda
Ram

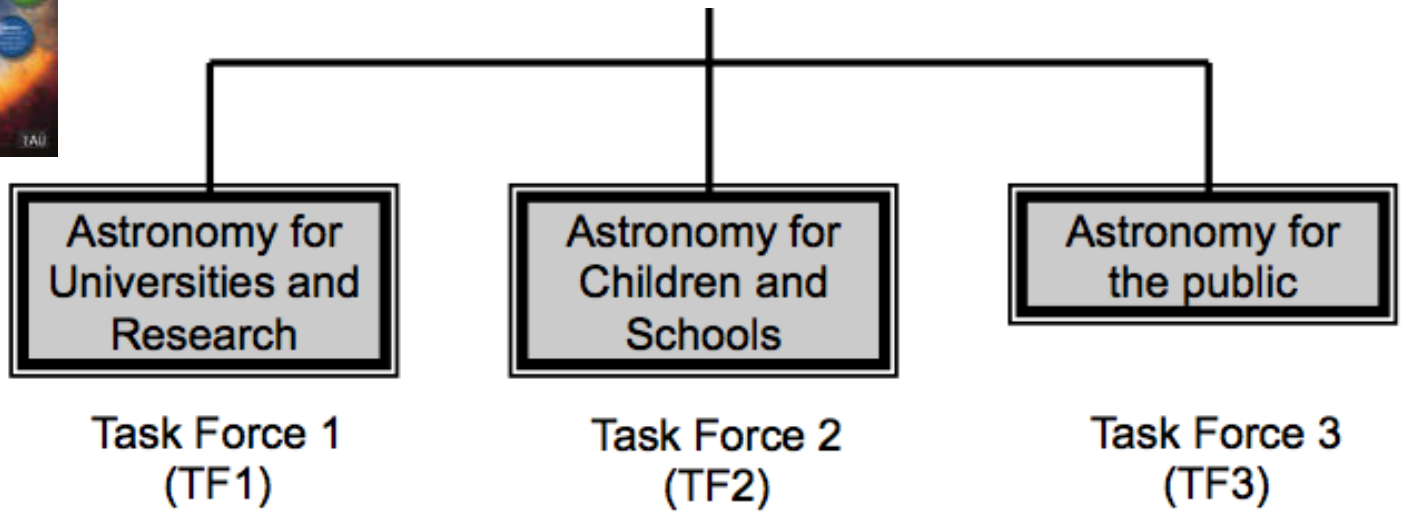
Interns:
Laure
Rajin
Eli Kasai







OAD Task Force membership



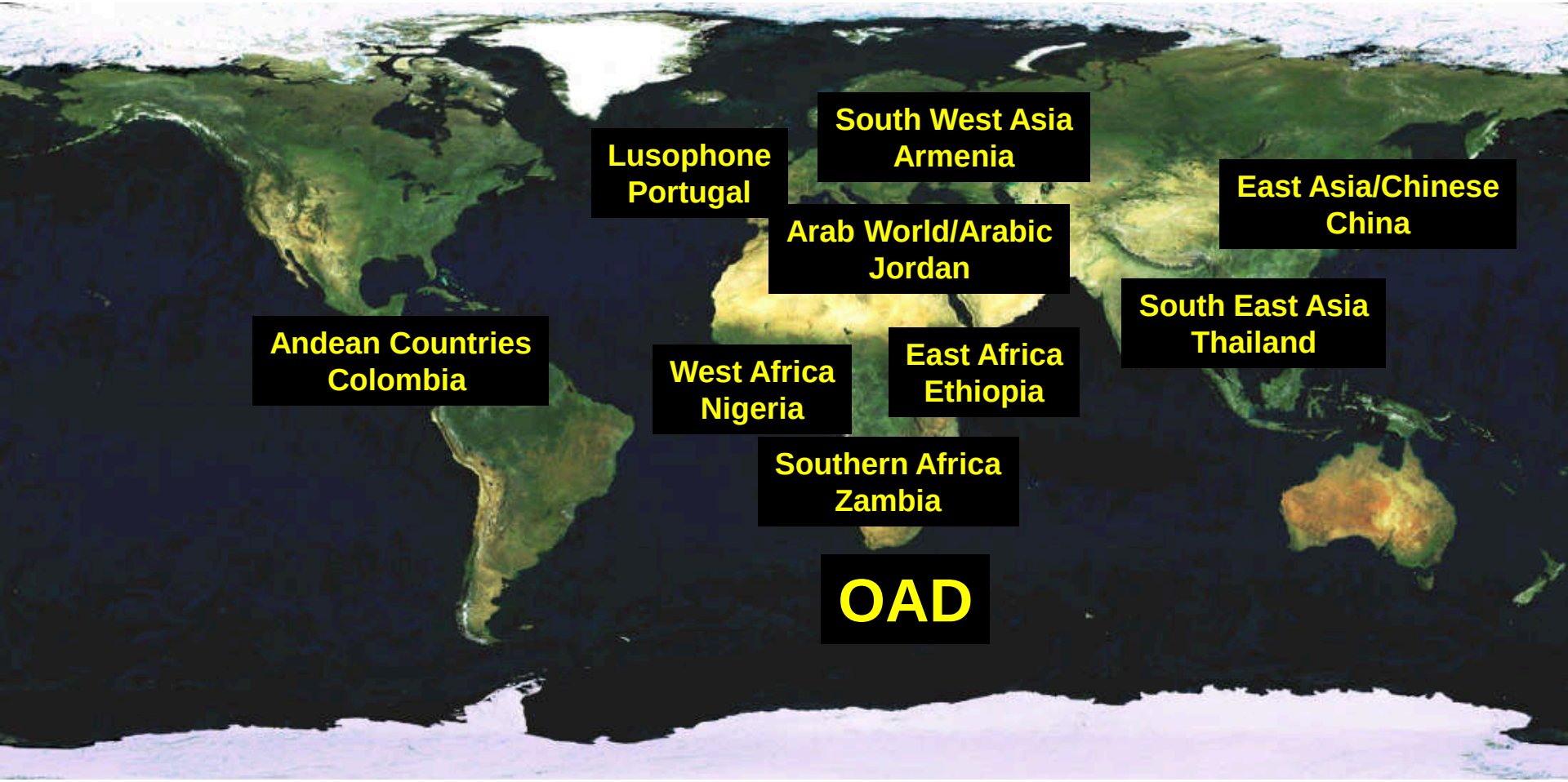
Jean-Pierre de Grève (Belgium, C46)
Richard de Grijs (China)
Michèle Gerbaldi (France)
Edward Guinan (USA – Chair)
Roger Hajjar (Lebanon)
Edward Jurua (Uganda)
Katrien Kolenberg (USA - VC)
Hakim Malasan (Indonesia, Div C)
Shengbang Qian (China)
Nicole van der Bliek (Chile)

Rosa Doran (Portugal)
Edward Gomez (LCOGT,Cardiff - VC)
Mary Kay Hemmenway (USA, **observer**)
Robert Hollow (Australia)
Ofodum Chukwujekwu Nworah (Nigeria)
Rosa Maria Ros (Spain)
Pedro Russo (Leiden/UNAWE/C55, Chair)
Cecilia Scorza (Venezuela/Germany)
Linda Strubbe (Canada)
Akihiko Tomita (Japan)
Jinhua He (China, Observer)

Thilina Heenatigala (Sri Lanka)
Sarah Kendrew (UK - New Media)
Kimberley Kowal (US)
Carolina Ödman (South Africa – VC)
German Puerta (Colombia)
Sze-leung Cheung (Japan, OAO)
Ian Robson (UK - Chair)
Komiko Usida (Japan)
Ziping Zhang (China)



OAD Regional Offices



OAD Volunteers





- IAU members, amateurs, professionals, teachers, students, public
- Over 550 worldwide (on this map they are grouped by location)

www.astro4dev.org/volunteers

OAD Partners

- Royal Astronomical Society
- Netherlands Organisation for Scientific Research
- International Centre for Theoretical Physics
- Inter-University Centre for Astronomy and Astrophysics
- University of Central Lancashire
- Haus der Astronomie
- Fiat Physica
- Associated Universities Inc. / Leiden University

Funded Projects (68+18=86)



www.astro4dev.org/proposals

TASK FORCES

- ASTRONOMY FOR CHILDREN & SCHOOLS (TF2)
- ASTRONOMY FOR UNIVERSITIES & RESEARCH (TF1)
- ASTRONOMY FOR THE PUBLIC (TF3)



OAD Annual Call for Proposals

	2012 Call (191)	2013 Call (230)	2014 Call (131)	2015 Call (124)
Astronomy for Universities and Research	42 applications € 752,959 requested	54 applications € 919,308 requested	31 applications € 514,103 requested	28 applications € 232,257 requested
Astronomy for Children and Schools	96 applications € 772,079 requested	113 applications € 864,731 requested	67 applications € 431,695 requested	66 applications € 654,802 requested
Astronomy for the public	53 applications € 310,782 requested	63 applications € 453,805 requested	33 applications € 178,254 requested	30 applications € 301,431 requested
Total requested (recommended)	€ 1,835,820 (€ 869,426)	€ 2,237,844 (€1,534,513)	€ 1,124,052 (€535,373)	€ 1,188,490



ASTRONOMÍA PERIFÉRICA
arte y astronomía en Bogotá

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erimentos espaciales (espacio)























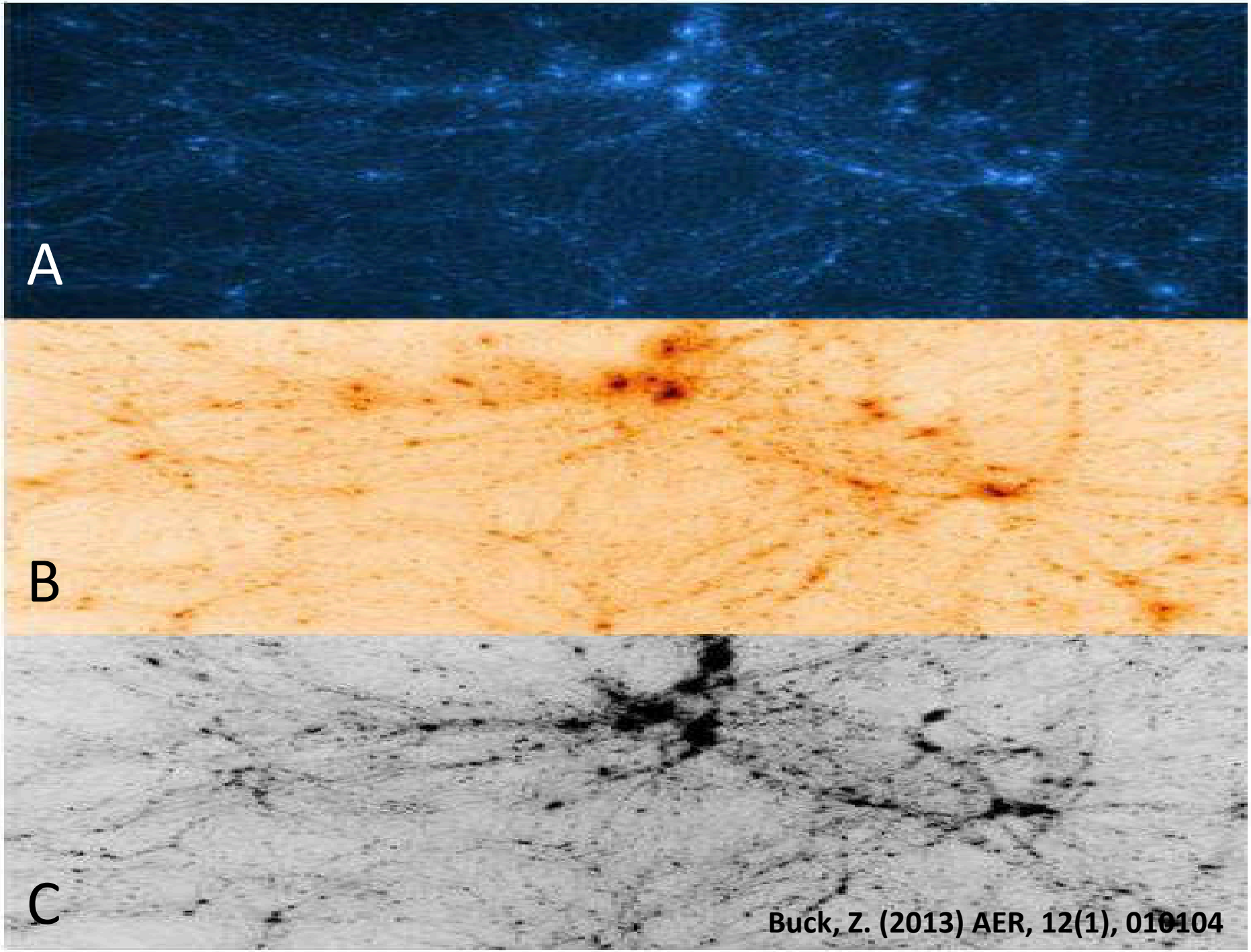


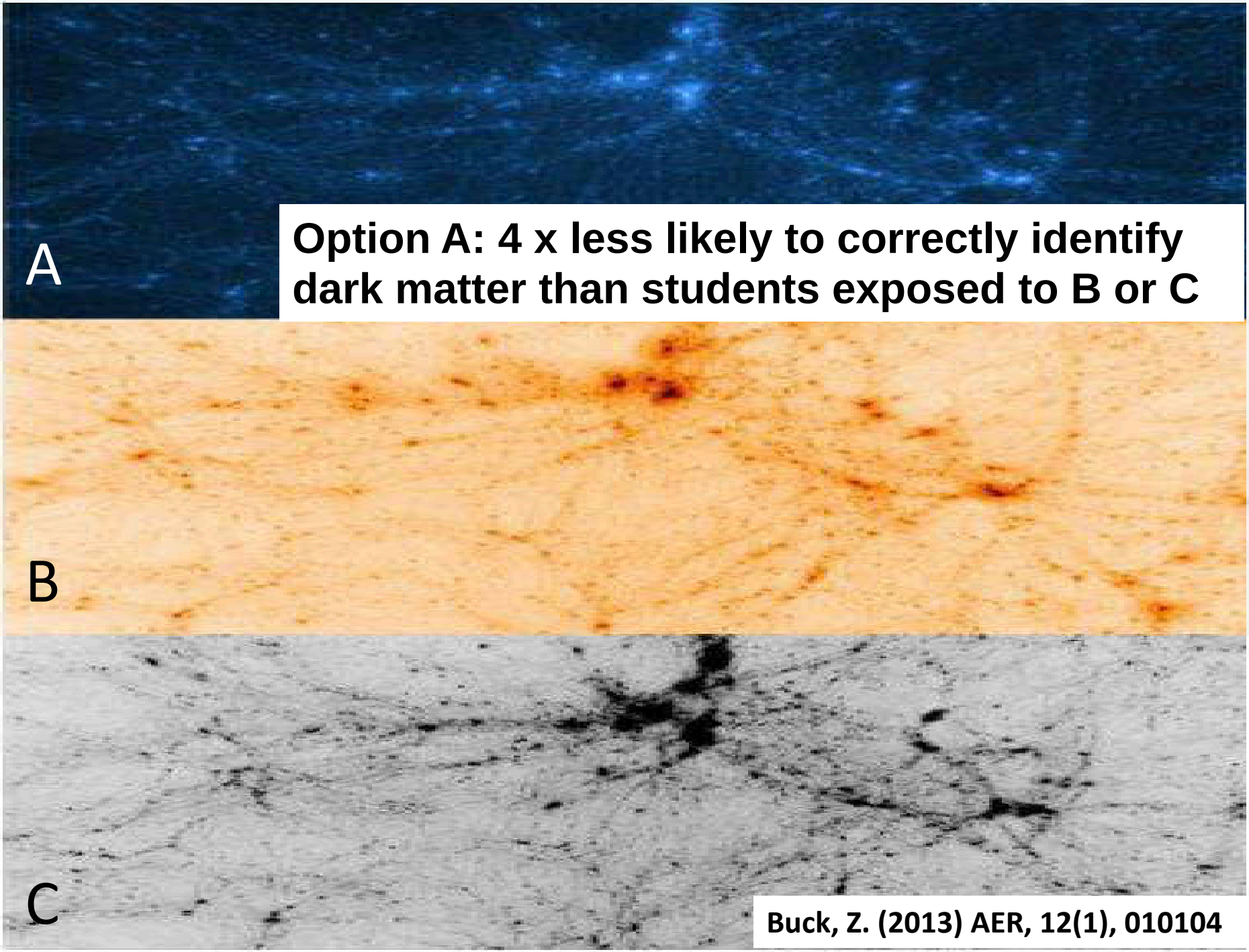


Do projects work?

- Access ≠ Development
- Humans are complex and embedded in complex social systems
- OAD needs to
  - Identify best practices, allocate resources efficiently
  - Manage risks of unintended consequences




A
B
C
Buck, Z. (2013) AER, 12(1), 010104


A
Option A: 4 x less likely to correctly identify dark matter than students exposed to B or C
B
C
Buck, Z. (2013) AER, 12(1), 010104

Best Evidence
Encyclopedia
Empowering Educators with Evidence on Proven Programs

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Search

Program Reviews

Mathematics
- Elementary
- Middle/High School
- Effectiveness of Technology

Reading
- Beginning
- Upper Elementary
- Elementary
- Middle/High School
- English Language Learners
- Struggling Readers
- Effectiveness of Technology

Science
- Elementary
- Secondary (New!)

Comprehensive School Reform
- Elementary (CSRQ)
- Middle/High School (CSRQ)
- K-12 Meta-Analysis (Borman)
- Education Service Providers (CSRQ)

Early Childhood
- Early Childhood Education (New!)

Methods
- Methodological Features and Effect Sizes (New!)

New Reviews Added to the BEE!

The BEE has recently added two major new reviews:

Secondary Science. A comprehensive review of research on science programs for grades 6-12.

Early Childhood Education. A comprehensive review focusing on studies comparing programs for four-year-olds using either "balanced" approaches, which include phonemic awareness and early phonics along with traditional preschool activities, to "developmental" approaches, which include little focus on pre-reading skills.

Other reviews being substantially updated and revised include:

- Elementary math
- Secondary reading
- Methodology effects in systematic reviews

Watch this space!

Spotlight

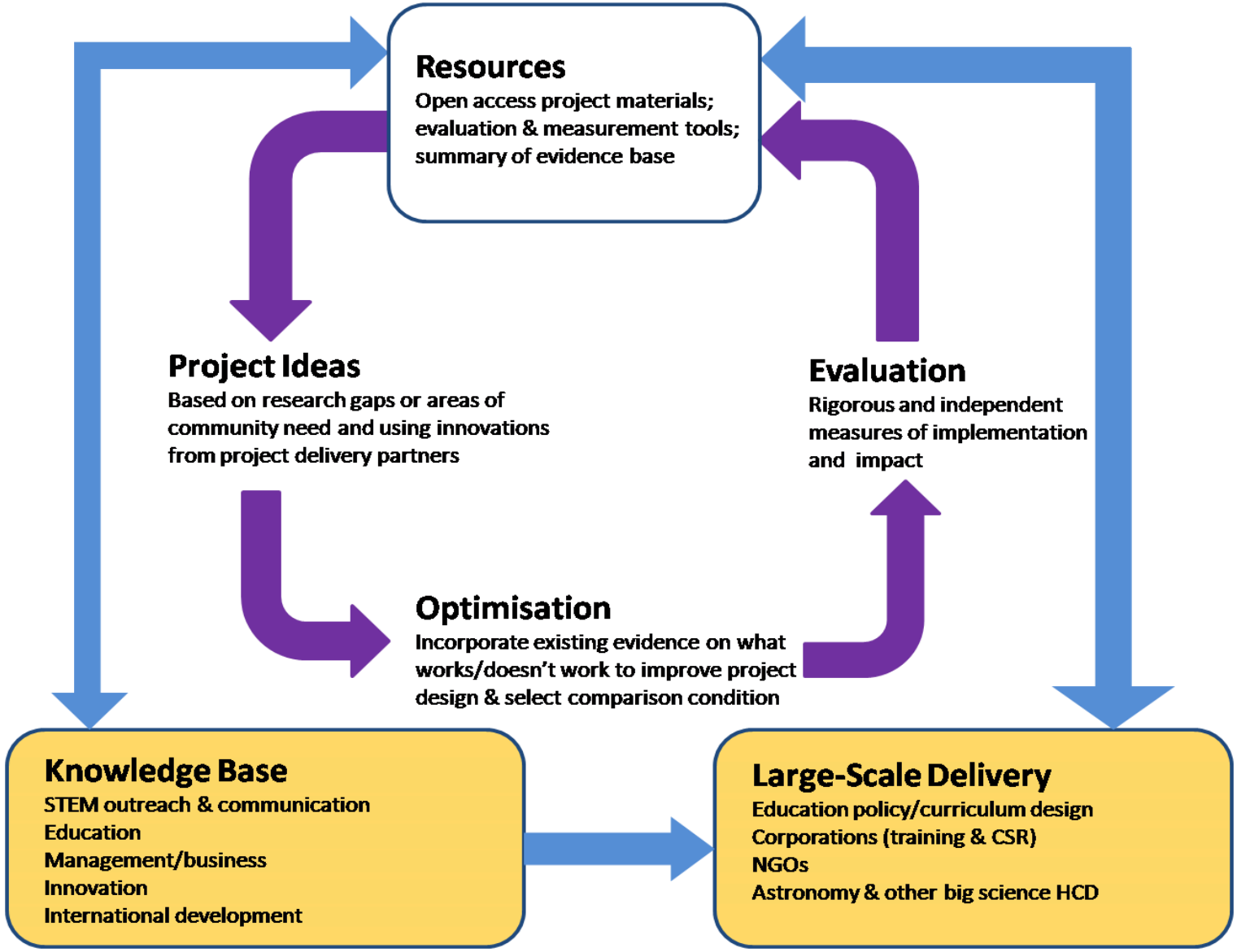
Effective Programs for Secondary Science



Our new review summarizes the evidence on four types of programs designed to improve the science achievement of students in grades 6-12

www.bestevidence.org

OAD Impact Cycle: Positive Feedback Loop



IAU GA 2015 Resolution

(following positive OAD review)

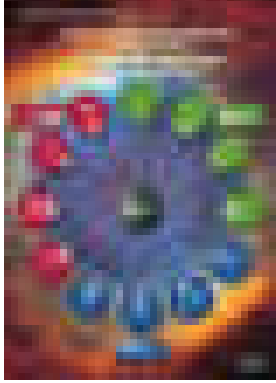


…

Resolves

1. That the pursuit of the goals of the Strategic Plan: Astronomy for the Developing World should continue until the XXXI General Assembly to be held August 2021,
2. That the Executive Committee should present for approval at the XXX General Assembly to be held in Vienna, Austria in August 2018 an extended Strategic Plan which addresses the future of the OAD and its activities beyond 2021,
3. That the Executive Committee should consult existing and potential stakeholders in the preparation of this Strategic Plan.

A possible vision for 2021



- User-friendly “impact cycle” (positive feedback)
  - Strategic partnerships with development organisations
- Synergised regional leadership
- Host large externally funded programmes
- Adoption by other sciences
- Volunteer system
- Renewed Strategic Plan

Shared resources/services



- Websites
- Mailing lists
- Evaluation
- Surveys
- Language
- Lobbying
- Virtual meetings
- Shared information on big astronomical events
- Certification
- Involvement with OAO in Japan

Common Plan across Regions

- Guiding Principles
- 2021 Strategic Goals
- Operational Objectives and Indicators
- Major Challenges
- Risk analysis
- Human Resources and Infrastructure
- Annual Budgets

Face to Face Meeting: 2-4 March 2016



Astronomy for a Better World!

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www.astro4dev.org

References

- Haynes et al. (2009). A surgical safety checklist to reduce morbidity and mortality in a global population. *The New England Journal of Medicin*e, *360*, 491–499.
- Yeager and Walton (2011) Social-Psychological Interventions in Education: They're Not Magic. Review of Educational Research, 267-301.